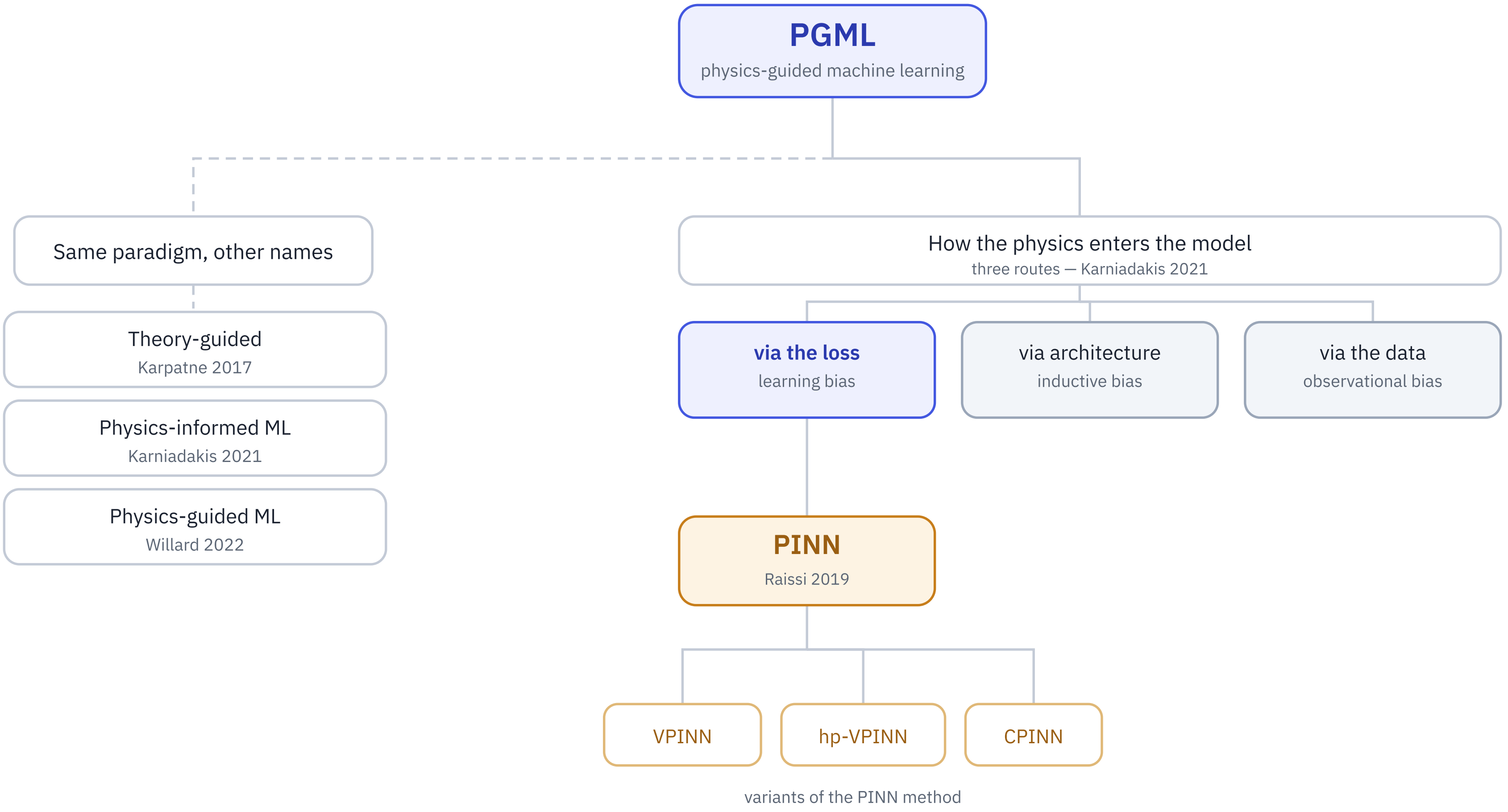
PGML and PINN sit at different levels

PGML is the family of ML models constrained by physics. PINN is one method inside that family.

One umbrella, several names, several methods



umbrella / loss route · PINN & variants · dashed line = the same paradigm under other names

SECTION 2.4 · THE TWO DIFFERENTIATORS

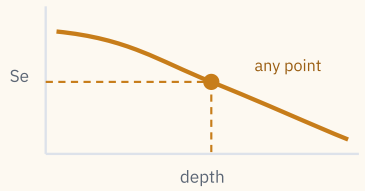
Same loss family

01

Shape of the output

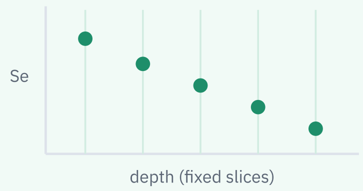
continuous vs discrete

PINN



one smooth curve, read at any point

This work



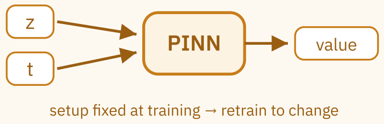
values on a fixed grid

02

What the network is

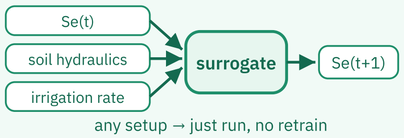
solution vs operator

PINN



the answer to one setup

This work



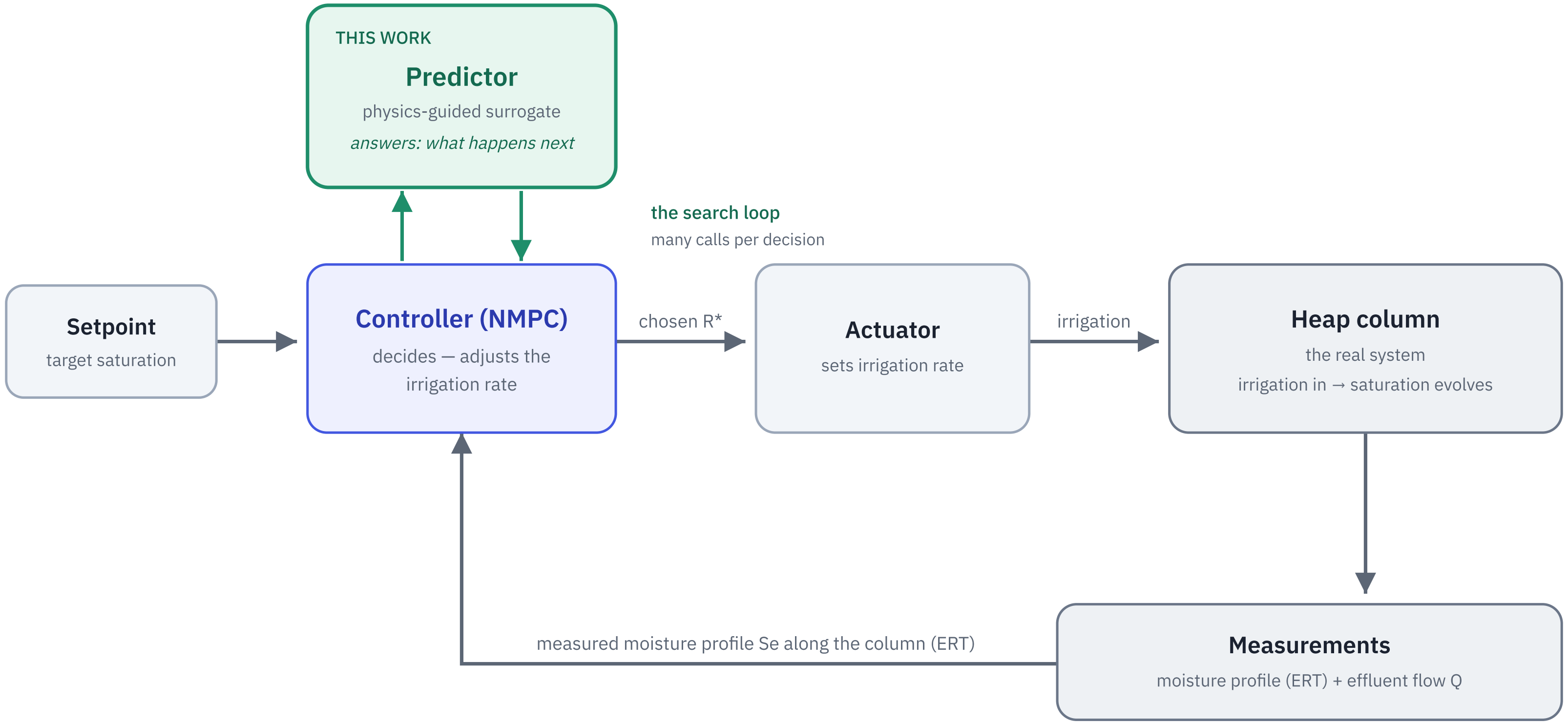
a calculator for any setup

PINN and this work: same loss, different object

	PINN	This work
WHAT IT LEARNS	The answer to one setup	A calculator: state now → state next
FORM OF THE ANSWER	A smooth curve, readable at any point in space and time	Values on a fixed grid, stepped forward in time
HOW PHYSICS GOES IN	Through the loss, as a residual penalty `same in both`	
CHANGE THE SOIL OR IRRIGATION	Retrain the network	Just run it, within its trained range
RUNS LIVE IN A CONTROLLER	No	Yes

A PINN learns the answer. This work learns the calculator that makes the answers.

Where the model sits



Inner loop — the search

The controller adjusts the irrigation rate and calls the predictor to test each try, repeating until the predicted next profile approaches the setpoint.

Outer loop — the beat

One decision per control step, hourly or daily. That step length is the sampling period. Sense, decide, act, then sense again.

The predictor is this work

It replaces the per-step identification and forward solve inside the NMPC (Olivares 2025). It never decides. It only answers "what happens next," fast.

The five papers this framing rests on

1 Karpatne, A. et al. (2017). Theory-guided data science: a new paradigm for scientific discovery from data. IEEE Transactions on Knowledge and Data Engineering, 29(10), 2318–2331.
`umbrella — theory-guided`

2 Karniadakis, G. E. et al. (2021). Physics-informed machine learning. Nature Reviews Physics, 3, 422–440.
`umbrella + PINN as a member (three routes)`

3 Willard, J. et al. (2022). Integrating scientific knowledge with machine learning for engineering and environmental systems. ACM Computing Surveys, 55(4), Article 66.
`umbrella — physics-guided, method classes`

4 Cuomo, S. et al. (2022). Scientific machine learning through physics-informed neural networks: where we are and what's next. Journal of Scientific Computing, 92(3), 88.
`nomenclature + PINN as one of a family`

5 Raissi, M., Perdikaris, P. & Karniadakis, G. E. (2019). Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics, 378, 686–707.
`the PINN definition`